TITANIC DATA ANALYSIS

Comprehensive Statistical & Visualization Report

Total Passengers	Survivors	Survival Rate	Average Age	Average Fare
891	342	38.4%	29.4	$32.2

Prepared by: Naveed Ahmad

Project: Titanic Data Analysis Project

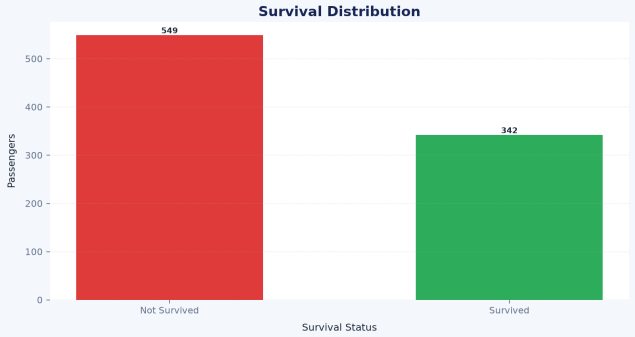
Charts: 28 Unique Visualizations

Technology: Python, Pandas, NumPy, Matplotlib, ReportLab

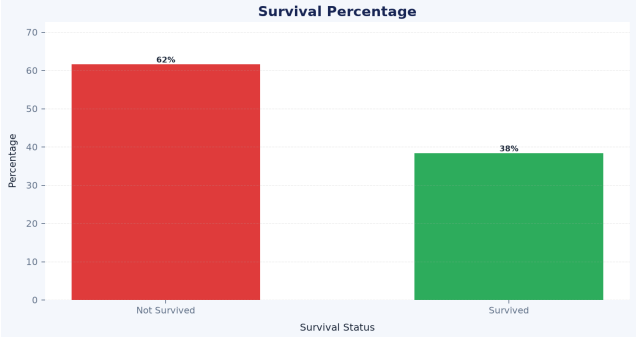
1. Executive Summary

The Titanic dataset contains **891** passenger records. Among these passengers, **342** survived while **549** did not. The overall survival rate was **38.4%**. The average passenger age was **29.4** years and the average fare was **$32.2**.

The analysis examines survival through demographic, class, age, family, fare, embarkation, and multivariate perspectives. The report contains 28 unique visualizations with an interpretation provided below each chart.



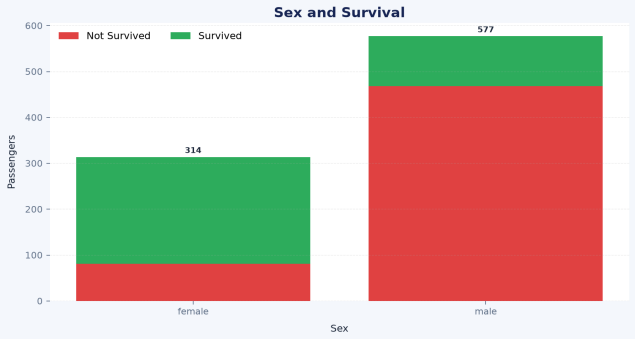
Interpretation: The chart shows 342 survivors and 549 passengers who did not survive.



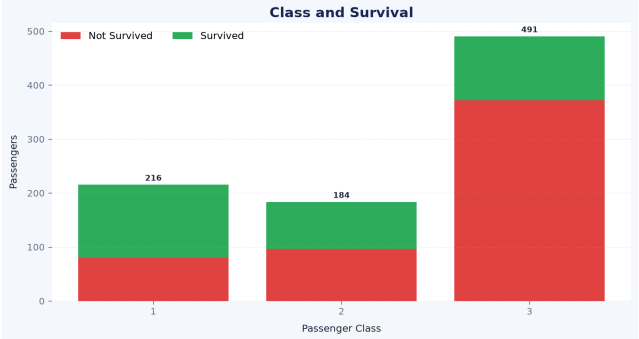
Interpretation: The overall survival percentage was 38.4%, while the remaining 61.6% of passengers did not survive.

2. Survival Analysis

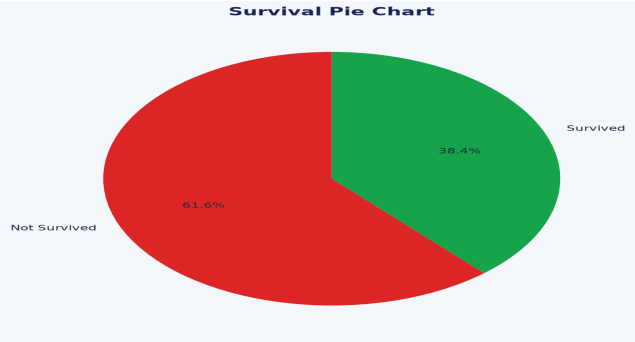
This section examines the survival outcome and compares survival patterns across key passenger characteristics.



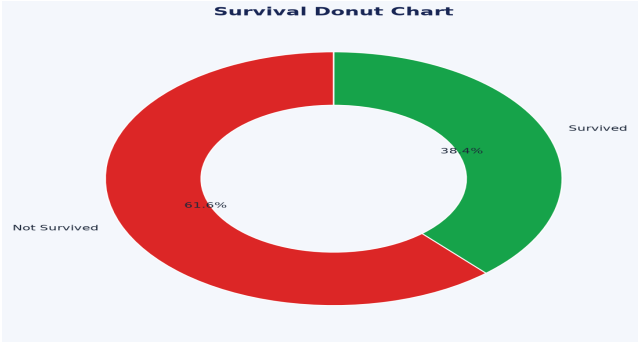
Interpretation: Female passengers had a survival rate of 74.2%, compared with 18.9% for male passengers.



Interpretation: Survival rates by class were 63.0% for Class 1, 47.3% for Class 2, and 24.2% for Class 3.



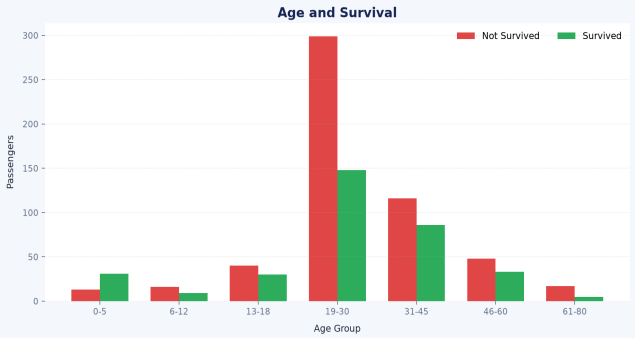
Interpretation: The pie chart represents the overall survival composition. Survivors account for 38.4% of the dataset and non-survivors account for 61.6%.



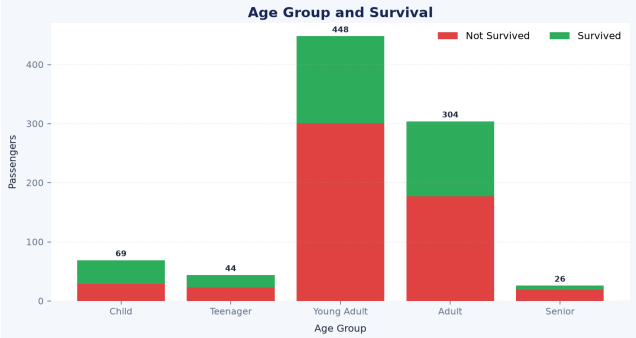
Interpretation: The donut chart provides the same survival composition in a compact form: 38.4% survived and 61.6% did not survive.

3. Demographic Analysis

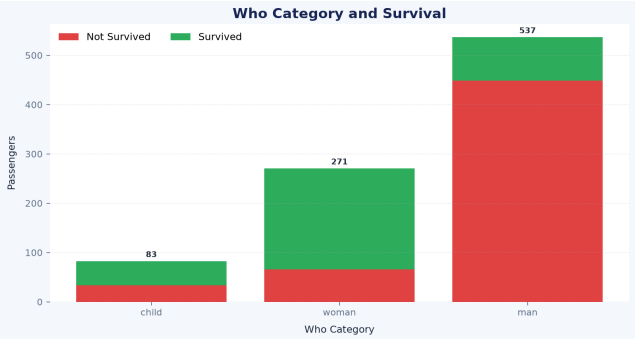
Demographic variables provide a detailed view of passenger composition and survival patterns across age and sex categories.



Interpretation: Survival varied across the defined age intervals. The chart allows passenger survival counts to be compared across children, teenagers, young adults, and older groups.



Interpretation: The age-group comparison shows variation in survival proportions across the passenger age categories. Child: 58.0%, Teenager: 47.7%, Young Adult: 32.8%, Adult: 41.8%, Senior: 26.9%.



Interpretation: The Who-category chart compares survival among children, women, and men. Child: 59.0%, Woman: 75.6%, Man: 16.4%.



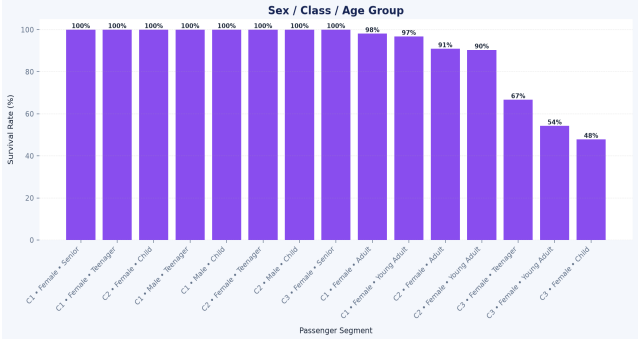
Interpretation: The largest sex-and-class combination was 3 / male, containing 347 passengers.

4. Passenger Class and Age Analysis

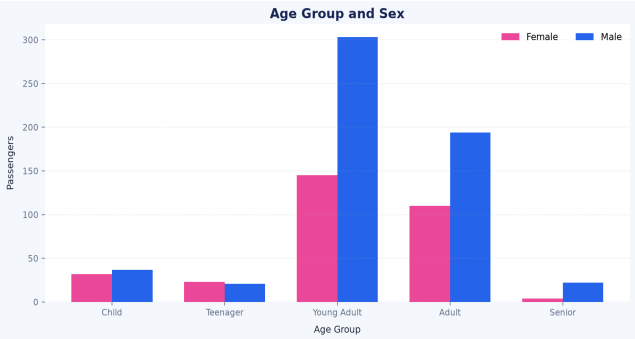
Passenger class, age, and sex are combined to examine passenger composition and segment-level survival patterns.



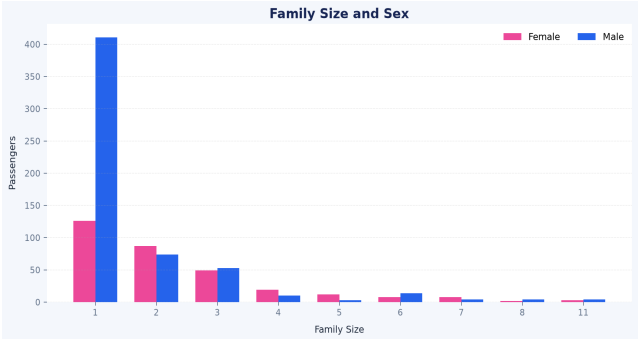
Interpretation: The most common class-and-age-group combination was 3 / Young Adult, with 299 passengers.



Interpretation: The multivariate chart shows how survival changes when class, sex, and age group are considered together. Among segments with at least five passengers, 1 / female / Teenager recorded 100.0% survival (7 passengers).



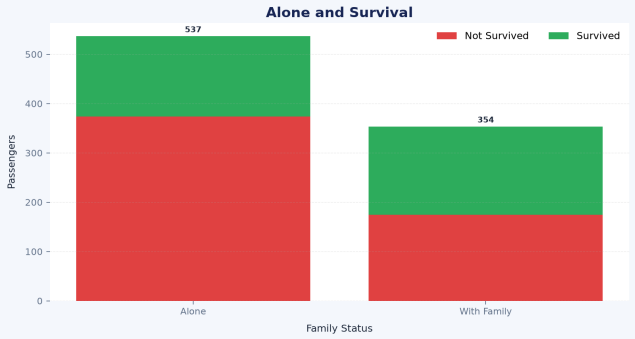
Interpretation: The largest age-group and sex combination was Young Adult / male, containing 303 passengers.



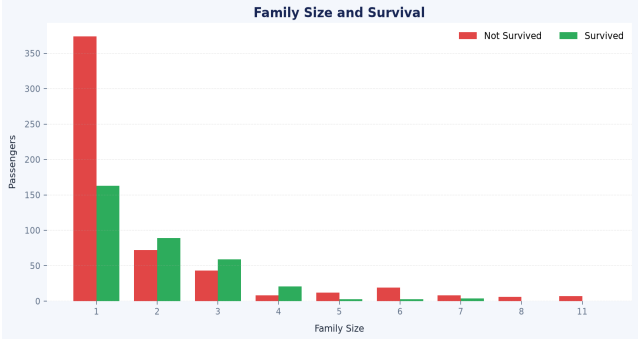
Interpretation: The largest family-size and sex combination was 1 / male, with 411 passengers.

5. Family Structure Analysis

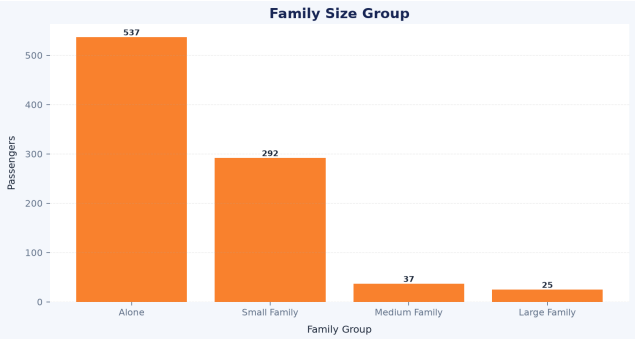
Family-related variables are derived from SibSp and Parch and are used to study family size and survival.



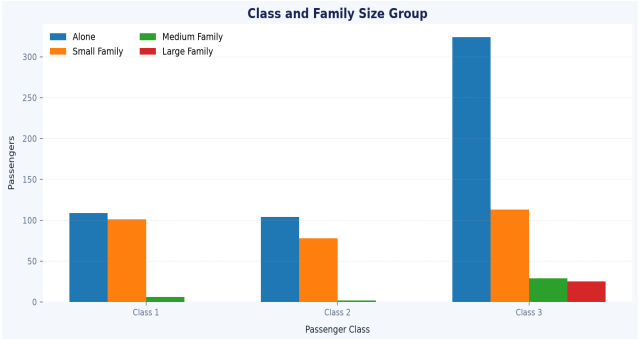
Interpretation: Passengers travelling alone had a survival rate of 30.4%, while passengers with family had 50.6%.



Interpretation: Survival rates varied by family size. Among family sizes represented by at least five passengers, the highest observed rate was 72.4% for family size 4 (29 passengers).



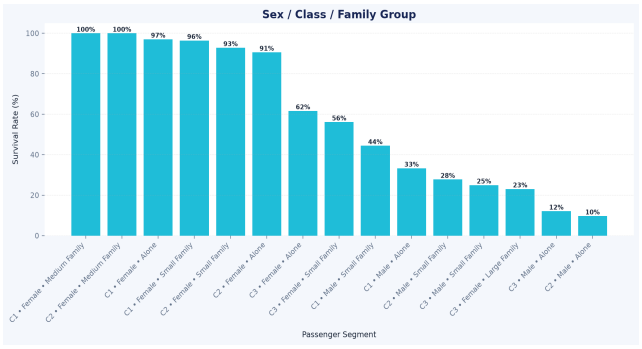
Interpretation: Alone was the most common family category, containing 537 passengers.



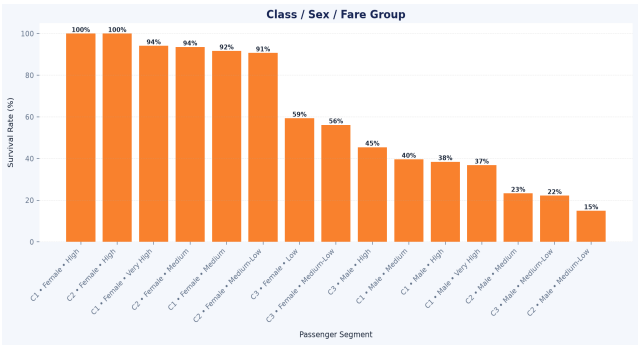
Interpretation: The most common class-and-family-group combination was 3 / Alone, with 324 passengers.

6. Multivariate Family Analysis

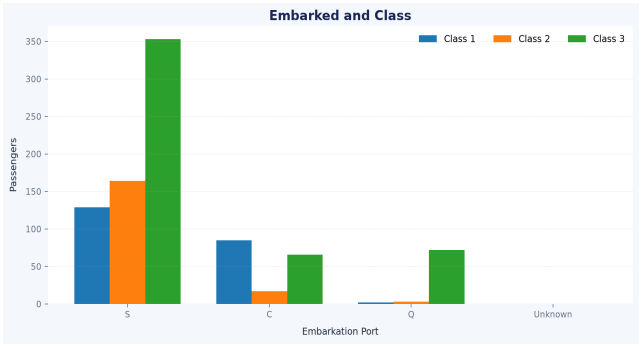
Multivariate charts combine several categorical variables to show more detailed passenger segments and their survival proportions.



Interpretation: Combining class, sex, and family group reveals additional survival variation. The highest rate among segments with at least five passengers was 97.1% for 1 / female / Alone (34 passengers).



Interpretation: The combined class, sex, and fare analysis shows differences between passenger segments. Among segments with at least five passengers, 1 / female / High had a survival rate of 100.0% (48 passengers).



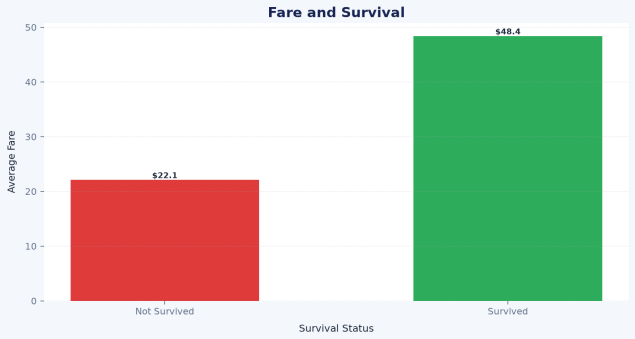
Interpretation: The most common embarkation-and-class combination was S / 3, with 353 passengers.



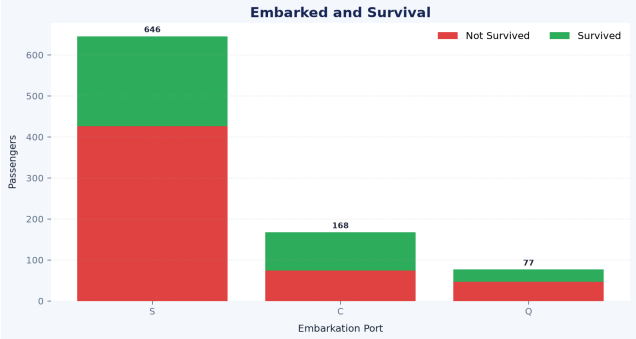
Interpretation: The largest sex-and-fare-group segment was male / Low, with 272 passengers.

7. Fare and Embarkation Analysis

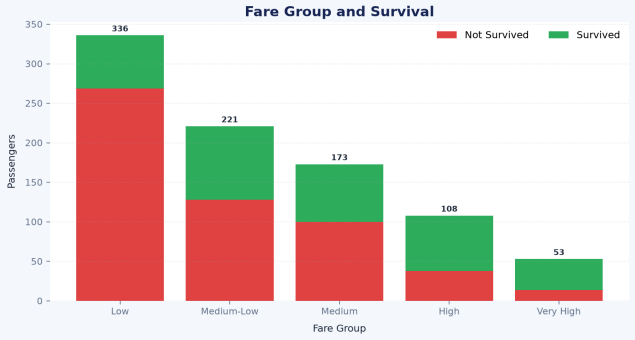
Fare groups and embarkation ports are used to examine passenger segmentation and survival.



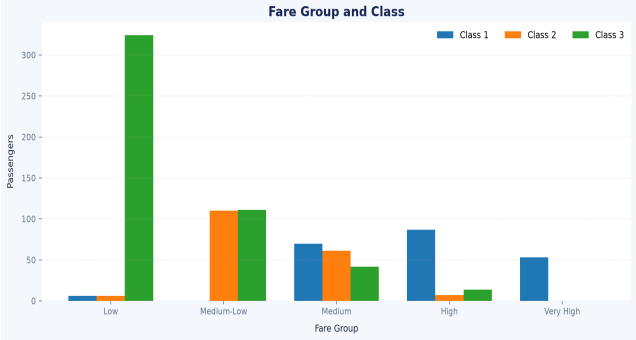
Interpretation: The average fare was $48.4 among survivors and $22.1 among passengers who did not survive.



Interpretation: Survival differed across embarkation ports. S: 33.9%, C: 55.4%, Q: 39.0%.



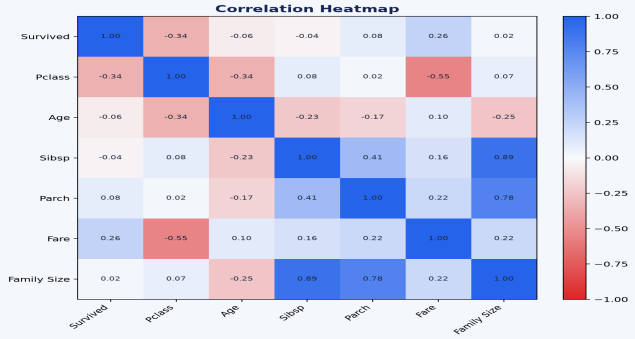
Interpretation: Survival proportions varied across fare groups. Low: 19.9%, Medium-Low: 42.1%, Medium: 42.2%, High: 64.8%, Very High: 73.6%.



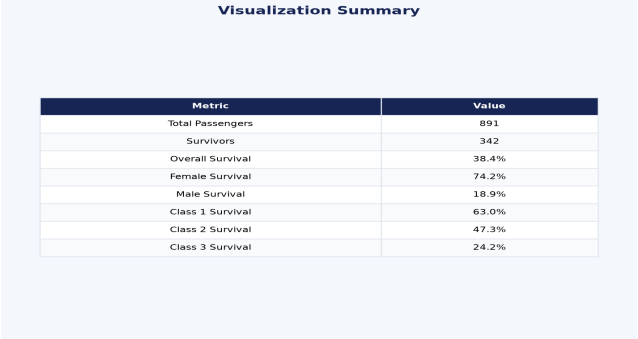
Interpretation: The largest fare-group and class combination was Low / 3, containing 324 passengers.

8. Correlation and Statistical Overview

The final statistical section presents the correlation structure and a consolidated visualization summary.



Interpretation: The strongest absolute linear correlation with survival among the included variables was with pclass (r = -0.34). Correlation describes linear association and does not by itself establish causation.



Interpretation: The summary table consolidates the main descriptive results: 891 passengers, 342 survivors, and an overall survival rate of 38.4%.

9. Analytical Observations

Area	Observation
Overall Survival	The overall survival rate was 38.4%.
Sex	Female survival rate was 74.2%, while male survival rate was 18.9%.
Passenger Class	Class 1: 63.0%; Class 2: 47.3%; Class 3: 24.2%.
Average Age	The average passenger age was 29.4 years.
Average Fare	The average fare was $32.2.

10. Methodology and Tools

Data Preparation: Dataset cleaning, numeric conversion, missing-value handling, and derived-variable creation.

Derived Variables: Family Size, Alone, Age Group, Family Group, Fare Group, and Who Category.

Analysis: Descriptive statistics, group comparisons, survival-rate calculations, cross-tabulation, and correlation analysis.

Visualization: Bar charts, stacked charts, percentage charts, pie chart, donut chart, and correlation heatmap.

Reporting: Professional dashboard and analytical report generated using ReportLab.

11. Conclusion

The Titanic analysis demonstrates how passenger-level data can be transformed into meaningful statistical and visual insights. The project integrates data preparation, descriptive analysis, categorical comparisons, family and age segmentation, fare analysis, multivariate visualization, and correlation analysis. The resulting dashboard provides a compact executive view, while the final report documents the analytical process and presents interpretations for all 28 visualizations.

Developed by Naveed Ahmad